

Harry Leung

Software Engineer

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EDUCATION

University of California, Irvine

Sep. 2021 – Jun. 2025

B.S. in Computer Science

- **Relevant Coursework:** Machine Learning, Neural Networks & Deep Learning, Artificial Intelligence, Graphical Models, Information Retrieval, Database Systems, Operating Systems, Data Structures & Algorithms, Networks

WORK EXPERIENCE

Software Development Engineer

Nov. 2025 – Present

Amazon Music

San Francisco, CA

- Shipped a candidate-model rollout framework: a parallel pipeline with Step Functions orchestration, scoped IAM, and isolated storage for production-readiness validation of a new entity scoring and clustering model; consolidated ~600 lines of duplicated infrastructure into shared parameterized constructs.
- Built backfill tooling (Lambda, SQS, Glue) that regenerates ML features at catalog scale to complete the feature store for the candidate model, migrating inference from sync to async to handle volume; added a reporter routing missing-input items to the Mozak ML feature-generation pipeline.
- Built the offline model evaluation layer (Glue/Athena dashboards, batch accuracy reports, cross-account analytics) and a data-quality AI agent comparing candidate-vs-production outputs with cross-script (CJK/Latin) false-positive detection and multi-batch aggregated reporting.
- Stood up ~30 CloudWatch alarms across SageMaker endpoints (memory, CPU, backlog age, invocation failures) and queue infrastructure; right-sized inference instances and concurrency to deliver ~24× per-instance throughput and eliminate cold-start timeouts.

Software Engineering Intern

Jun. 2023 – Sep. 2023

Teradyne Inc. (Litepoint Division)

San Jose, CA

- Built a Tkinter-based GUI for data visualization and regression analysis, reducing new-user training time from 2 hours to under 45 minutes (~60% reduction).
- Migrated regression test data storage to Apache Cassandra, reducing average query response time by ~95% across 100+ GB datasets.
- Integrated Matplotlib reporting into the analysis workflow, surfacing failure patterns that previously required manual log inspection and accelerating root-cause investigation for the QA team.

PROJECTS

Image Safety RAG | Python, RAG, OpenCLIP, FastAPI, Faiss, MCP

[GitHub](#)

- Built an explainable image content-moderation system: OpenCLIP ViT-B/32 embeddings indexed in HNSW (Faiss/Weaviate), retrieved via FastAPI, packaged into MCP context manifests, and reasoned over by a multimodal LLM (GPT-4o or local Phi-3-vision).
- Grounded moderation decisions in retrieved nearest-neighbor exemplars from the LAION-Safety subset, enabling the LLM to cite concrete prior cases when classifying and explaining new uploads.

Toy Language Interpreter & Compiler | Python, Compilers, LLVM IR, AST

[GitHub](#)

- Designed a programming language from scratch: custom lexer, recursive-descent parser, AST, and tree-walking interpreter supporting variables, arithmetic, control flow, and recursive functions with error reporting.
- Extended the toolchain with an LLVM IR code-generation backend, lowering the interpreted language to a compilable target with explicit type handling.

SKILLS

Languages: Python, TypeScript, Go, C++, Java, SQL, Rust

AWS / Cloud: CDK, Lambda, Step Functions, SageMaker, Glue, Athena, SQS, S3, CloudWatch, IAM

ML / Data: PyTorch, Hugging Face Transformers, NumPy, pandas, scikit-learn, Keras

Web & Frameworks: React, Next.js, Node.js, GraphQL, REST

Systems & Tools: Docker, Kubernetes, Linux, Git, Redis, MySQL, Bazel, CMake, GDB, Valgrind

Spoken: English, Cantonese, Mandarin